

Split Flap Display Firmware

Firmware for a 12-module Chainlink/Denver3D-style split-flap display (ESP32), including the self-hosted WiFi web UI (message, presets, automation, timer).

What you need

- **PlatformIO** — install the [PlatformIO IDE extension](#) for VS Code (free).
- **USB cable** for your ESP32 (the T-Display board used here).
- This folder, unzipped, opened in VS Code.

Setup

1. Open this folder in VS Code (PlatformIO will auto-detect `platformio.ini`).
2. In `firmware/esp32/splitflap/`, copy `secrets.h.example` to a new file named `secrets.h`, and edit it:
 - `WIFI_SSID` / `WIFI_PASSWORD` — your WiFi network
 - `DEVICE_INSTANCE_NAME` — the mDNS name for the display (e.g. `splitflap` gives you `splitflap.local`)
 - Leave `MQTT_*` and `OTA_PASSWORD` as-is unless you plan to use MQTT or wireless (OTA) updates — see comments in the file for details.
3. `secrets.h` is required for the project to build — it's excluded from this package on purpose since it holds credentials.

Build & upload

1. Connect the ESP32 via USB.
2. In VS Code's PlatformIO sidebar, select the **chainlink** environment (already the default — builds for 12 modules with the web UI enabled).
3. Click **Upload** (or from a terminal: `pio run -e chainlink --target upload`).
4. Open the Serial Monitor to confirm it boots. Once connected to WiFi, the display's screen shows its IP address — open that address (or `http://<DEVICE_INSTANCE_NAME>.local/`) in a browser for the web UI.

More info

- `firmware/esp32/splitflap/WEB_UI_README.md` — details on the wireless web UI (message/presets/automation/timer), known limitations, and how it stores data.
- `firmware/esp32/README.md` — how the ESP32 build layers on top of the shared splitflap driver code.
- `platformio.ini` — all build environments. `chainlink` is what you want for a standard build; `chainlink_ota` is the same build but uploads wirelessly once you've flashed once over USB. The `advanced_*` environments are for different hardware (Chainlink base station, driver tester) — skip those unless you know you need them.

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Apache License 2.0 — see `LICENSE.txt`.